NORTH SOUTH UNIVERSITY

Department of Electrical & Computer Science

Project Report: ULTRASONIC RADAR SYSTEM

Course Code : CSE323

Section : 09

SUBMITTED BY

SUBMITTED TO

Student Name & ID:

Course Instructor: Mua3

Faisal Mahmud Abir | 2132358642

ULTRASONIC RADAR SYSTEM

Faisal Mahmud Abir, Undergraduate Student, Department of Electrical and Computer Engineering, North South University, Dhaka , Bangladesh

***Abstract*—Radio detection and ranging (RADAR) systems use ultrasonic waves to detect physical parameters like distance, speed, position, range, direction, and size of both stationary and moving objects. This technology is used in a variety of sectors, including military installations and commercial applications. Radartechnology has come a long way, especially in the field of navigation. This study examines current navigational technology and suggests a RADAR system based on Arduino. By using less power and being compatible with a variety of Arduino programmers and open-source code, this system has advantages over conventional RADAR systems. A servo motor with an ultrasonic sensor placed on it rotates at predetermined angles and speeds as part of the system. Both the ultrasonic sensor and the servo motor are connected to the Arduino's digital input/output pins, enabling efficient and accurate detection.**

Keywords : Arduino, Ultrasonic sensor, Servo motor, Radar

I. INTRODUCTION

Ultrasonic is a non-contact level measurement method that uses sound waves to determine the process material being measured. Ultrasonic transmitters operate by sending a sound wave, generated from a piezoelectric transducer, to the media being measured. The device measures the length of time it takes for the reflected sound wave to return to the transducer. A successful measurement depends on reflection from the process material in a straight line back to the transducer. However, various influences affect the return signal. Factors such as dust, heavy vapors, tank obstructions, surface turbulence, foam, and even surface angles can affect the returning signal. That is why the conditions that determine the characteristics of sound must be considered when using Ultrasonic measurement.

A. OBJECTIVES :

The aim of this project is to design, develop, and implement an Ultrasonic Radar System for distance measurement and object detection. The primary objectives include :

Accurate Distance Measurement : Develop a system that accurately measures the distance between the radar device and the surrounding objects using ultrasonic waves.

Object Detection and Tracking : Implement a mechanism to detect and track objects within the radar's range, providing real-time data on their movement.

Multi-Device Compatibility : Ensure compatibility with various devices or displays, allowing users to monitor the radar system remotely through a centralized interface. It is optional because it depends on user preference.

Customizable Alerts and Notifications : Implement a notification system to alert users in case of predefined events, such as the presence of an object within a specified proximity.

Energy Efficiency : Optimize the system's power consumption to extend its operational life and provide a sustainable long-term solution for distance measurement and object detection.

B. Scope

There are a lot of scopes to develop our project. If we can use wireless Ultrasonic sensors it is possible to cover a full surrounding area. We can also see the object that come in the coverage area of radar by using a camera

II. IMPLEMENTATION

A. Hardware system design for Arduino1

Hardware system consist of basically 3 components named as Arduino, servo-motor, and ultra-sonic sensor. Ultrasonic sensor is mounded upon a servo motor which helps it to move and provide it a turning mechanism. Both ultrasonic sensor and servo motor are controlled and powered by Arduino. As given in above figure 1 we can see both ultrasonic sensor and servo motor is powered by Arduino.

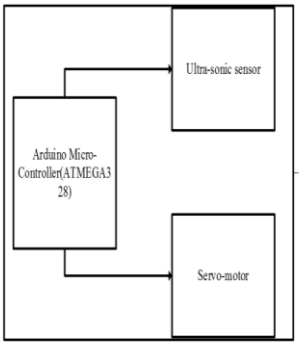


Figure 1. Hardware System Design of Radar System.

B. System circuit design

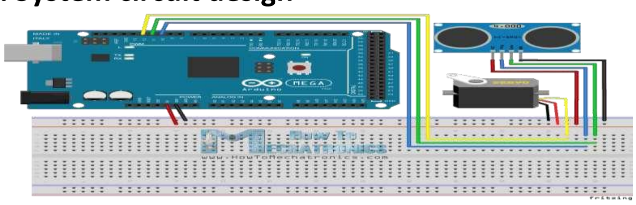


Figure 2

Figure shows hardware system design which was designed using fritzing environment. It shows the connection of different electronics components. In the figure triggering pins of ultrasonic sensor is connected to D8 pin of Arduino, control line of servo motor is connected to D6 pin of Arduino and D7 pin of Arduino is connected to echo pin. VCC pins of servo motor and ultrasonic sensor is connected to 5V pin of Arduino while ground pin of Arduino is connected to ground pin of both servo motor and ultra-sonic sensor.

C. System circuit implementation on bread board –

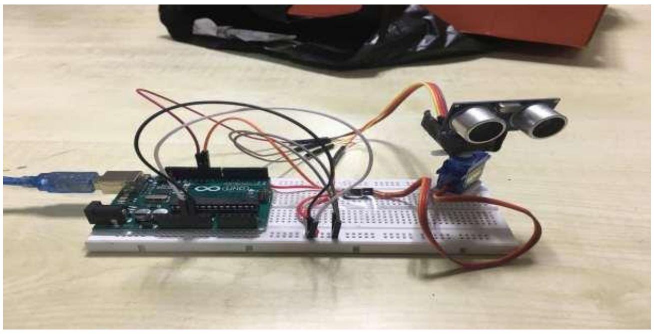


Figure 3. Breadboard of the hardware system implementation.

Above figure 3 shows complete implementation of hardware system. It can be seen that ultrasonic servomotor is placed upon a servo motor and it is placed above bread board. Arduino is placed in breadboard in other side of the breadboard and entire connection is made between them. Arduino and servo motor are stick to breadboard to stop it from tripping over when servo motor moves. Arduino IDE was used to write code and upload it in Arduino. Arduino code reads position of servo motor and calculate distance of nearest object in the path.

D. Hardware system testing –

A cable was used for connecting Arduino to develop developing machine. From Arduino IDE helped us to obtain result in serial monitor.

E. GUI system design and implementation :

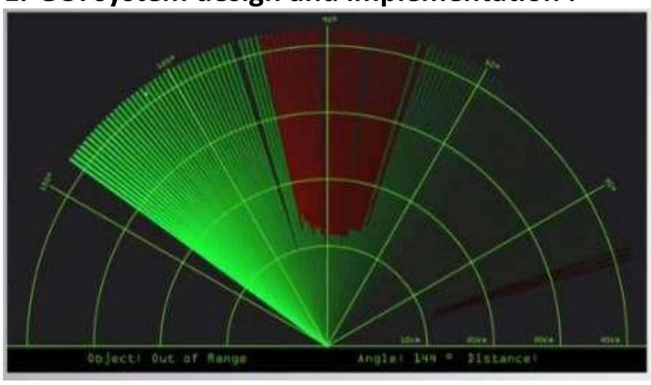


Figure 4. GUI Implementation for the mapping interface.

Object class of radar project represent object that it encounters such as distance, target/range and angle/direction of position of object. Distance () method (), angle () method, location () method takes the required value such as distance, angle and makes them on GUI to do simulation.

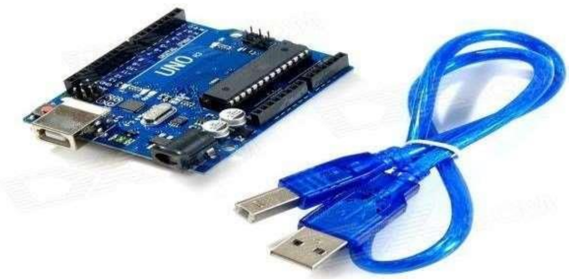
III. METHOLOGY

A. Working

The aim of this project is to calculate the distance position and speed of the object placed at some distance from the sensor.Ultrasonic sensor sends the ultrasonic wave in different directions by rotating with help of servo motor. This wave travels in air and getsreflected back after striking some object. This wave is again sensed by the sensor and its characteristics is analysed and output is displayed in screen showing parameters such as distance and position of object.Arduino IDE is used to write code and upload coding in Arduino and helps us to sense position of servo motor and posting it to the serial port along with the distance of the nearest object in its path. The output of sensor is displayed with the help of processing software to give final output in display screen.

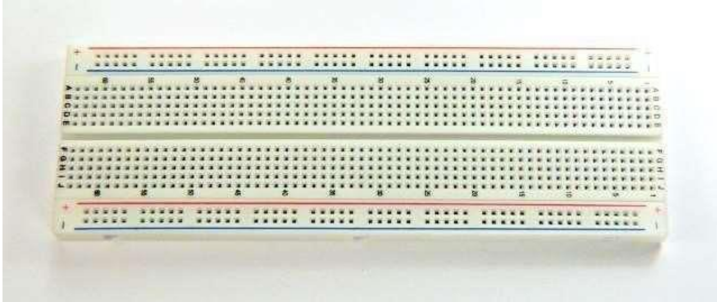
B. Hardware description

I) Arduino



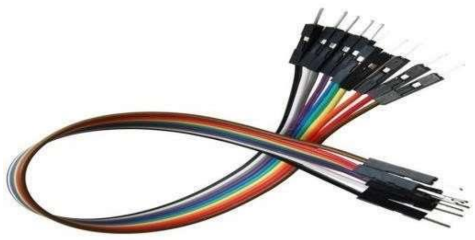
II) Ultrasonic sensor

III) Bread board



IV) Servo motor

V) Jumper wires



IV. RESULT AND DISCUSSION

A.Performance Testing

Implementation the hardware setup was completed using the components listed. The Arduino was programmed to rotate the servo motor incrementally, measure distances, and visualize the results on an LCD display.
Preliminary testing of the system in controlled environments yielded the following results:

- **Accuracy:** Distance measurements were accurate within ±1 cm for ranges up to 40cm.
- **Angular Resolution:** The system achieved a resolution of 1° with the servo motor.

• **Response Time:** The system processed data under 200 MS per scan cycle.

B.Limitations and Challenges

• **Range**: The detection range is limited to 40 cm, restricting its application in large-scale environments.

• **Environmental Sensitivity**: Performance is affected by soft surfaces, which absorb sound waves, reducing detection reliability.

V. ACKNOWLEDGEMENT

I would like to thank my professor Mosabber Uddin Ahmed for his expert advice and encouragement through this project. I would also like to thank my friends who always encouraged me throughout this period.

VI. CONCLUTION

In this paper a system radar system was designed with the help of Arduino, servomotor and ultrasonic sensor which can detect the position,distance of obstacle which comes in its way and converts it into visually representable form.This system can be used in robotics for object detection and avoidance system or can also be used for intrusion detection for location sizes.Range of the system depends upon type of ultra-sonic sensor used. We used HC-SR04 sensor which range from 2 to 40 cm.

VII. REFERENCES

[1] G. Bhor, P. Bhandari, R. Ghodekar and S. Deshmukh, "Mini Radar," International Journal of Technical Research and Applications, pp. 68-71, 2016.

[2] D. B. Kadam, Y. B. Patil, K. V. Chougale and. S. S. Perdeshi, "Arduino Based Moving Radar System," International Journal of Innovative Studies in Sciences and Engineering Technology (IJISSET), vol. 3, no. 4, pp. 23-27,2017.

[3] T. P. Rajan, K. K. Jithin, K. S. Hareesh, C. A. Habeeburahman and. A. Jithin, "Range Detection based on Ultrasonic Principle," International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, vol. 3, no. 2, pp. 7638-7643, 2014.

[4] P. S. Abhay,. S. K. Akhilesh, P. Amrit and Kriti, "A Review on Ultrasonic Radar Sensor for Security system," Journal ofEmerging Technologies and Innovative Research (JETIR), pp.137-140, 2016.

[5] S. Bharambe, R. Thakker, H. Patil, K. M. Bhurchandi,"Substitute eyes for blind with navigator using android", Proc.Texas Instrum. India Edu. Conf. (TIIEC), pp. 38-43, Apr.2013.

[6] D. Sunehra, A. Bano, S. Yandrathi, "Remote monitoring and control of a mobile robot system with obstacle avoidance capability", Proc. Int. Conf Adv. Comput. Commun. Inf.(ICACCI), pp. 1803-1809, Aug. 2015.

[7] http://www.instructables.com/id/ATMega328-using-Arduino-/

[8] "Compressive radio detection and ranging imaging" in IEEE

[9] Wikipedia.

!